UNIVERSITY OF ENGINEERING & TECHNOLOGY MARDAN

COMPUTER SCIENCE

SUBJECT: COMPUTER NETWORKS

SUBMITTED BY & TO

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SECTION : A

SEMESTER : 4th

SUBMITTED T0 : NAJEEB SIR

LAB # 1

TASK 1

What is the hostname of your computer. Write below.

Command Prompt

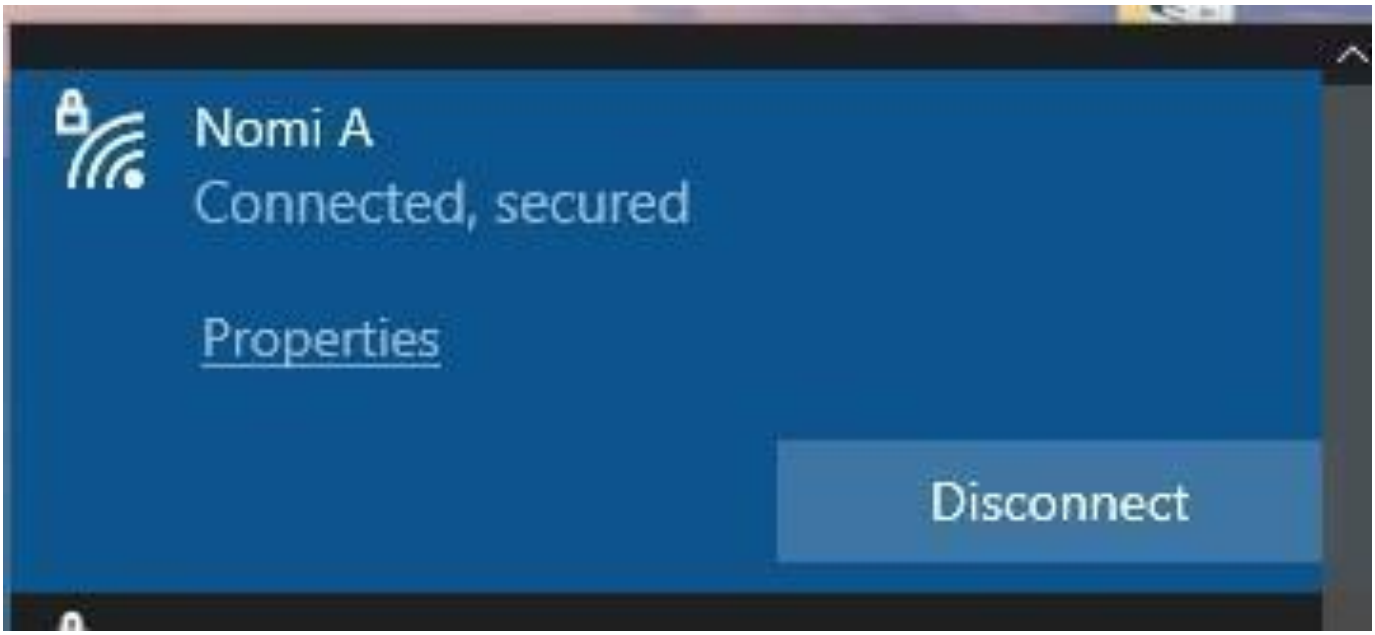
```
Microsoft Windows [Version 10.0.19045.5247]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Rifaq Ajmal>hostname
DESKTOP-2PPB8PP

C:\Users\Rifaq Ajmal>
```

TASK 2

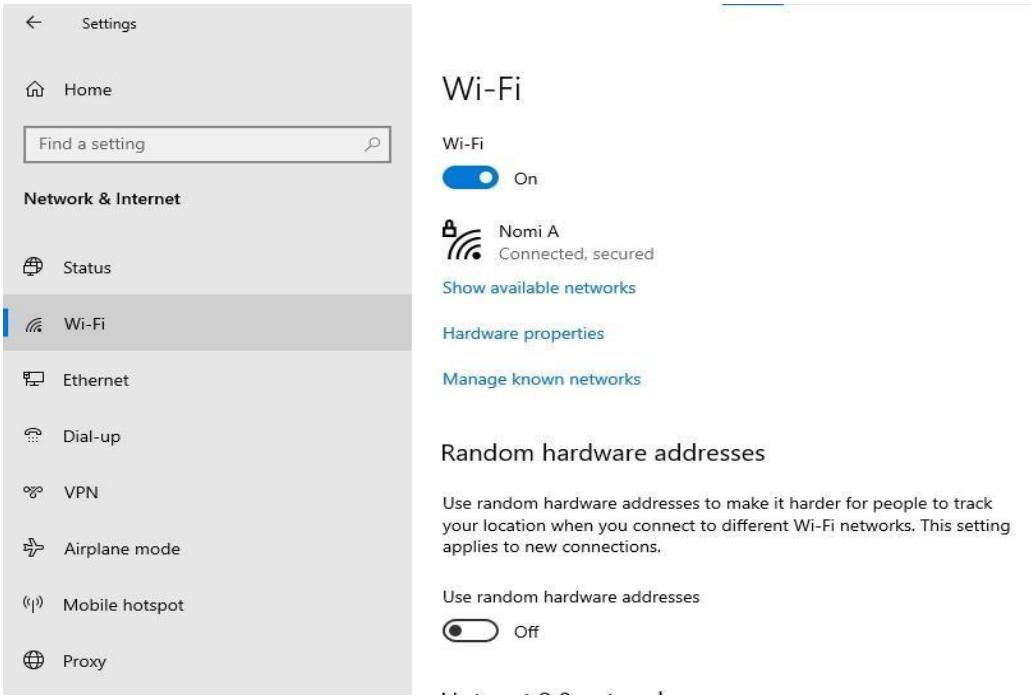
Find out which network you are connected to. Write the name of the connected network below.



TASK 3

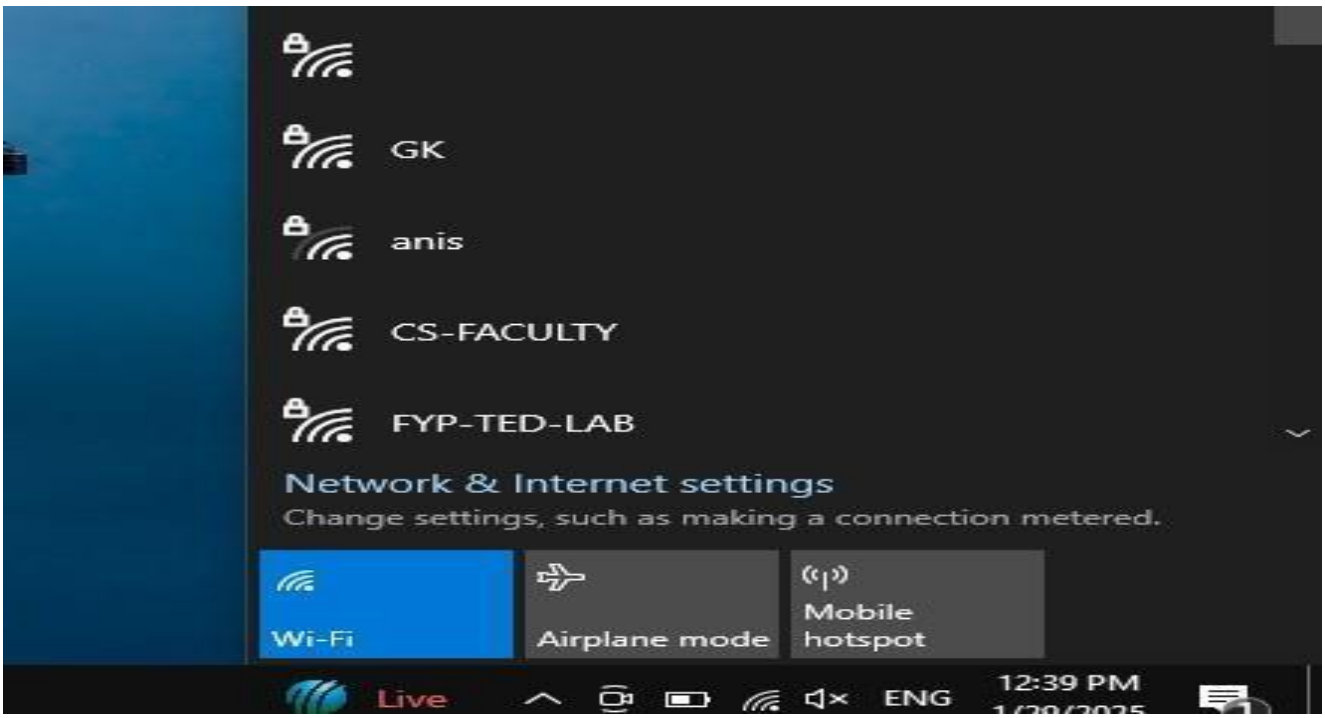
Is the network, you are connected to, wired or wireless?

ANS: The network we are connected is wireless….



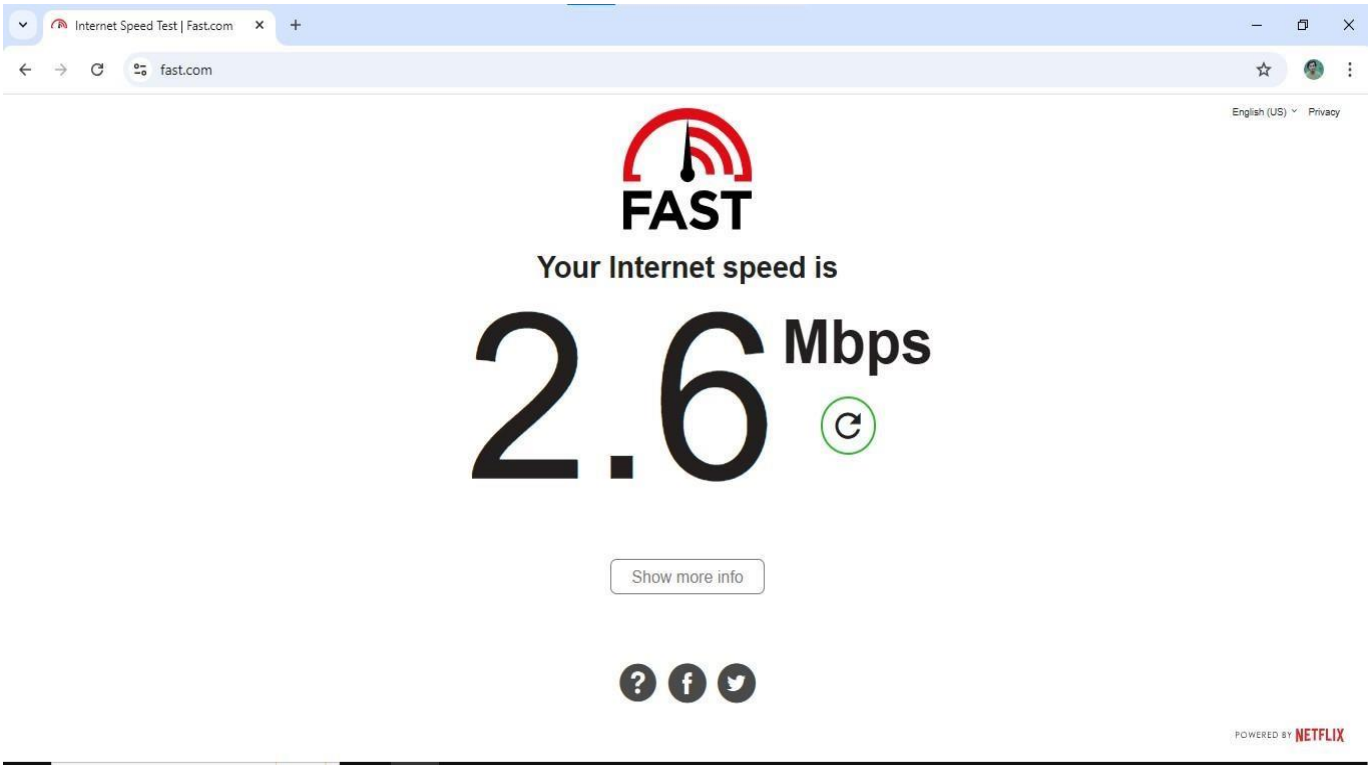
TASK 4

List the names of the networks available. Put a screenshot below.



TASK 5

Find out what speed you can get by using FAST.com. Put a screenshot below.



TASK 6

Find out the IP address of your computer. Put a screenshot below. ANS:

That is 192.168.248.17

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : 2402:ad80:107:4569:f232:899f:45a5:e9f8
   Temporary IPv6 Address. . . . . . : 2402:ad80:107:4569:5870:f163:5369:46c0
   Link-local IPv6 Address . . . . . : fe80::6e0a:2ae5:46b9:a0c9%20
   IPv4 Address. . . . . . . . . . . : 192.168.248.17
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
```

TASK 7

Find out the IP address of the DNS server. Put a screenshot below.

It Is using local router at IP 192.168.0.1 so that's why the DNS is not in 8.8.8.8 or 1.1.1.1

Select Command Prompt

```
Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : E6-A7-A0-95-89-C7
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Dual Band Wireless-AC 8260
   Physical Address. . . . . . . . . : E4-A7-A0-95-89-C7
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   IPv6 Address. . . . . . . . . . . : 2404:3100:1014:45c0:d7e2:20ea:109c:d2c9(Preferred)
   Temporary IPv6 Address. . . . . . : 2404:3100:1014:45c0:3041:569b:b729:8ec2(Preferred)
   Link-local IPv6 Address . . . . . : fe80::6e0a:2ae5:46b9:a0c9%20(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.60.16(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Wednesday, January 29, 2025 12:53:06 PM
   Lease Expires . . . . . . . . . . : Wednesday, January 29, 2025 1:55:07 PM
   Default Gateway . . . . . . . . . : fe80::fc74:96ff:fee1:a7f7%20
                                       192.168.60.76
   DHCP Server . . . . . . . . . . . : 192.168.60.76
   DHCPv6 IAID . . . . . . . . . . . : 350529440
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2E-BD-2E-CD-34-4B-50-00-00-00
   DNS Servers . . . . . . . . . . . : 192.168.60.76
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

TASK 8

Find out the MAC/physical address of the various network interfaces of your computer. Put a screenshot below.

Select Command Prompt

```
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 8:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : E4-A7-A0-95-89-C8
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

TASK 9

Find out the IP address of Google, Facebook, www.uetmardan.edu.pk and write in table below.

Website	IP Adress
Google	142.251.40.100
Facebook	157.240.241.35
www.uetmardan.edu.pk	166.62.27.63

TASK 10

For what purposes ping command can be used. Ping google.com and put a screenshot below.

Command Prompt

```
Microsoft Windows [Version 10.0.19045.5247]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Rifaq Ajmal>ping google.com

Pinging google.com [2a00:1450:4019:805::200e] with 32 bytes of data:
Reply from 2a00:1450:4019:805::200e: time=1099ms
Reply from 2a00:1450:4019:805::200e: time=653ms
Reply from 2a00:1450:4019:805::200e: time=415ms
Reply from 2a00:1450:4019:805::200e: time=583ms

Ping statistics for 2a00:1450:4019:805::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 415ms, Maximum = 1099ms, Average = 687ms

C:\Users\Rifaq Ajmal>
```

Purpose of using ping command

- **To check network connectivity.**
- **To measure latency.**
- **Troubleshoot network issues.**

  Verify DNS solution.
- **Test firewall settings**

THE END